

Design and Implementation of "Fighting and Dream" E-Sports Data Management System

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Abstract—With the rapid development of modern information technology, an industry named "E-sports" was born. The significance of this project lies in providing convenience for E-sports management through the Spring Boot, Vue.js and Mybatis development system. In this system, players can query event data or training arrangements, complete their training tasks in a timely manner, or provide feedback to coaches. Players need to check the official competition arrangements through the system to participate in the competition. Coaches can query the competition data and schedule of players, arrange training arrangements for each player in their team, or provide feedback to the organizers. The organizer can not only arrange the competition and publish it in the system, but also make modifications to various details such as the competition.

Keywords: Spring Boot, E-sports, data management, Vue.js, MyBatis

1. Introduction

1.1. Background and Significance of the Project

With the rapid development of mobile internet in today's world, people's entertainment methods have become diverse, shifting from traditional real-life entertainment to virtual entertainment on the internet. Due to the competitive nature of some entertainment projects, an industry called "E-sports" has emerged. With the advantages of the Internet's "fast dissemination" and "wide audience", this industry has gained considerable popularity among young people. The global esports industry continues to expand in scale and maintain a high growth rate, and will make its debut as an official event at the Hangzhou Asian Games. With domestic athletes achieving excellent results in one E-sports event after another, E-sports is increasingly recognized by more and more people. Both practitioners and spectators of E-sports related events are steadily increasing, which is an important factor for the future development of the E-sports industry.

Through this system website, one type of E-sports event data can be managed. The website users are mainly divided into four categories: athletes, coaches, event organizers,

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and administrators. Different users have corresponding different permissions, and the operations that can be performed with different permissions are different. The functions that these users can perform can meet the basic requirements of an esports data management system. For example, in addition to querying event data, the system also allows users with corresponding permissions to manage event data such as athletes and coaches. Event organizers can also arrange and publish events through the system.

As an emerging and rapidly developing industry, while taking advantage of the advantages of the Internet, it will also be affected by the negative side of the Internet. Some E-sports events have failed to effectively manage their competitors or organizations, resulting in scandals that are constantly detrimental to the development of the industry. With the help of the internet, E-sports events not only need to pursue the principles of fairness and justice like traditional sports events, but also constrain themselves and their organizations to establish a good image for the audience in society. One step in establishing a sound management system requires a suitable management system to assist in management. With appropriate management, the E-sports industry can develop healthily and comprehensively like crops in fertile soil. The significance of this system lies in providing convenience for E-sports management.

1.2. Research Contents

This project mainly studies the implementation of a data management website for MOBA type E-sports competitions. Vue combined with Spring Boot and MyBatis is a popular technology in the current B/S development model [1]. Implement various functions of the backend through the Spring Boot framework technology, using the Spring Boot microservice architecture for applications [2]. The Spring Boot framework integrates Spring and the Spring MVC framework [3]; The front-end website pages are created using the Vue.js framework, which is currently the most commonly used Java Script framework [4]; MySQL database provides data management services.

1.2.1 Player User

As a participant in the event, the event data reflects a portion of the player's performance, so players are not allowed to modify it on their own. In addition to basic functions such as modifying information, the player identity user in the system is only allowed to query event data, arrange events, or provide feedback to the coach. The following are some of the main functions of the player identity user in the system.

- a) Query event data, training arrangements, and competition arrangements
- b) Modify a portion of the current account user information
- c) Modify a portion of the current account user information
- d) View user information for the current account

1.2.2 Coach User

In addition to basic functions such as querying event data, the coach users in the system can also arrange for their team's players to undergo training or provide feedback to the event organizers as team representatives. The following are some of the main functions of coach identity users in the system.

- a) Query event data and arrangement
- b) Arrange or modify the training plan of our team's players
- c) Provide feedback or suggestions to the event organizers
- d) Modify a portion of the current account user information
- e) View user information for the current account

1.2.3 *Sponsor User*

The main responsibility of the organizers and users of the competition is to arrange the competition or discuss after understanding the feedback from each team. If they believe that the team's demands are reasonable, they can make corresponding modifications to the details of the competition, such as the competition date and address. The following are some of the main functions of the event organizer users.

- a) Arrange the competition and publish it in the system
- b) Modify the game schedule based on feedback from the team coach
- c) Modify a portion of the current account user information
- d) View user information for the current account

1.2.4 *Administrator User*

As the user with the highest privileges in the system, the administrator user is responsible for managing user roles in the system and organizing event data after the end of the event, and finally compiling it into the personal event data of participating characters such as players or teams. The following are some of the main functions of administrator users.

- a) Manage various users of the system
- b) Organize event data and compile it onto the data of each contestant or participating user
- c) View user information for the current account

1.3. *Contributions and Innovations*

This project mainly studies the implementation of a data management system for MOBA type E-sports competitions, which is applied to the emerging industry "E-sports" and provides convenience for event data management in the E-sports industry. In the system, there are channels between players, coaches, and event organizers to provide information feedback level by level. The event organizer can use this system to complete the arrangement and release of E-sports events. Coaches can use this system to develop training plans for each player in their respective teams.

2. Method

2.1. *System Architecture*

The framework technologies is shown in Fig.1, it used in the front-end of this system are Vue, Vue Router, Element UI, Axios, etc; The front-end page is created using HTML, CSS, and JavaScript languages. The front-end receives data and then accesses the back-

end interface through Axios requests. The backend uses Spring Boot framework technology, with the Controller layer obtaining parameters from HTTP requests, the Service layer implementing business logic, and the DAO layer accessing the MySQL database through Mybatis.

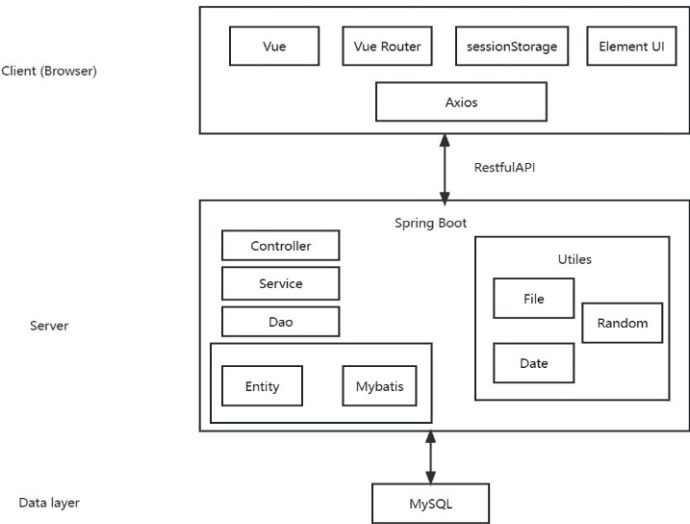


Figure 1. Architecture diagram

The users of the system are mainly divided into four types: athletes, coaches, administrators, and event organizers.

2.2. System functional structure

System functional structure is shown in Fig.2.

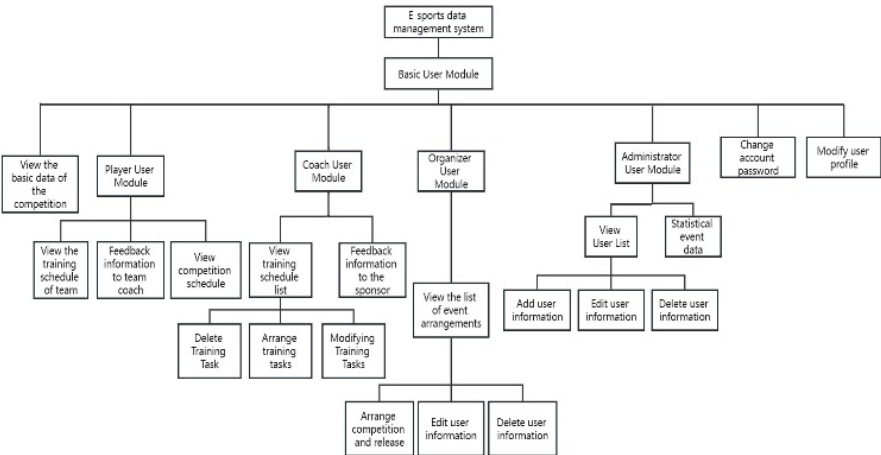


Figure 2. System functional structure

2.3. Introduction to System Technology

Vue is a progressive framework for building user interfaces, integrating HTML, CSS, and JavaScript, with the advantages of simplicity, lightness, and ease of use [5]. This framework can also display and reflect changes in data [6]. Spring Boot is a lightweight framework developed based on Spring, which not only has the excellent characteristics of Spring, but also has two core functions: control inversion and aspect oriented [7], and can solve the problem of difficult project resource management [8]. Spring Boot does not need to consider version compatibility and framework integration like Spring does [9]. MyBatis establishes a connection to the MySQL database according to service requirements and is responsible for executing specific business SQL statements and processing the execution results [10].

3. Experiment

The framework technologies used in the front-end of this system are Vue, Vue Router, Element UI, Axios, etc; The front-end page is created using HTML, CSS, and JavaScript languages. The front-end receives data and then accesses the back-end interface through Axios requests. The backend uses Spring Boot framework technology, with the Controller layer obtaining parameters from HTTP requests, the Service layer implementing business logic, and the DAO layer accessing the MySQL database through Mybatis.

3.1. Database Table Design

The database table design meets the connection between various entities and can meet the requirements for normal operation of system functions. Due to space limitations, here is a list of some table designs.

Table I is player table, which stores the identity information of player and event data. As the main subjects in the event, player data is the most important. The player data will be simply calculated to obtain a comprehensive data that will be displayed on the page as a simple reference.

TABLE I. PLAYER TABLE

field name	name	data type	constraint
pid	Player ID	int	Primary key, AUTO INCREMENT
pname	Player Name	int	not null
userid	Corresponding user number	int	Foreign Key
kills	Kill count	varchar(32)	
death	Number of deaths	int	
assist	Number of assists	int	
teamid	Team ID	int	FK

Table II is the competition schedule table, which stores the most important relevant information of a competition. The organizer will arrange the competition and display the relevant information for the relevant personnel to view. Relevant teams are required to participate in the competition on time.

TABLE II. COMPETITION SCHEDULE TABLE

field name	name	data type	constraint
gameid	Game ID	int	Primary key, AUTO INCREMENT
hometeam	Home team	int	Foreign Key
awayteam	Away team	int	Foreign Key
gamedate	RACE DATE	date	not null
gameaddress	Competition address	varchar(100)	not null

Table III is a user table used to store the identity and permission identifiers of each user role, as well as their login account, password, and other information.

TABLE III. USER TABLE

field name	name	data type	constraint
userid	User ID	int	Primary key, AUTO INCREMENT
lname	User login name	varchar(11)	not null
lpass	Password	varchar(11)	not null
username	Username	varchar(32)	not null
flag	Identity identification	int	Foreign Key
picture	User profile	varchar(100)	

3.2. Timing Diagram

Due to space constraints, a representative functional sequence diagram of the main roles in the system is presented here.

Fig.3 shows the user information modification function. The user management interface is only open to administrator users. After the administrator user enters the system, the system queries and displays the user information saved in the database. The administrator user can choose to edit, and after the modification is completed, a successful message will be returned.

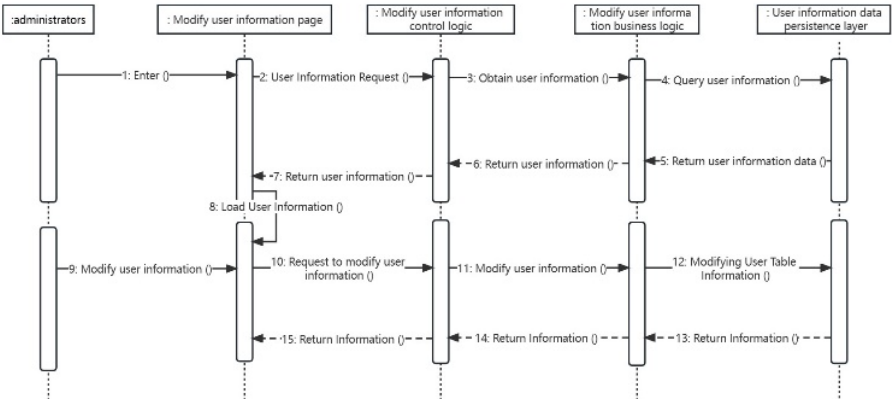


Figure 3. Function sequence diagram for modifying user information

After the coach user enters the training management page, the system will automatically query the team of the coach from the database. After querying the training arrangement information of the team, the coach user can choose to delete the record. The sequence diagram of this function is shown in Fig.4.

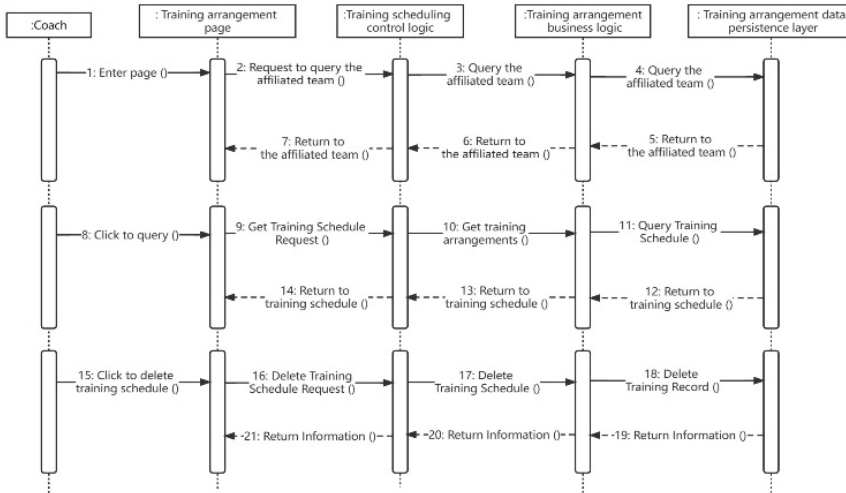


Figure 4. Coach Delete Training Arrangement Function Sequence Diagram

After entering the competition arrangement page, the organizer can edit the detailed information of the competition, such as the competition time and location. After the arrangement is completed, click the button to save the data to the database and publish it to the system. The process sequence is shown in Fig. 5.

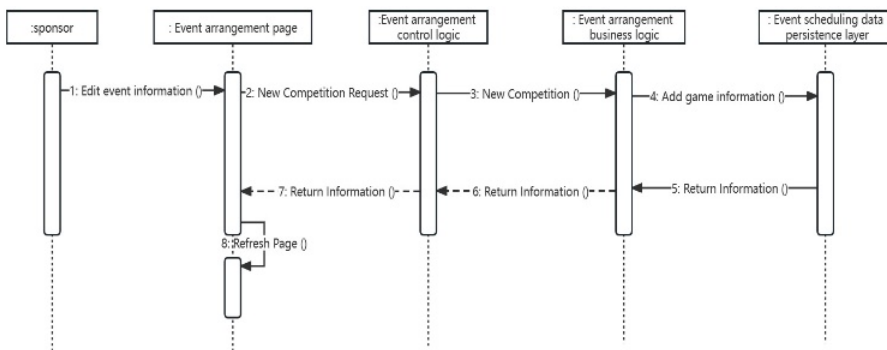


Figure 5. Sequence diagram of adding event functions by the sponsor

3.3. Quantitative Analysis

Quantitative analysis and simple testing are conducted to address the issue of system management data capacity. The quantitative analysis research scope included 50 players and 10 coaches to evenly allocate user roles to 10 teams. The system temporarily arranges

30 events through one event organizer and publishes them to the system. After the end of a competition, a player will receive three pieces of data: kill, assist, and death, and use these three pieces of data to calculate a comprehensive data. Therefore, after the end of a competition, a total of 40 data changes will be made for ten players from both sides. Each player provides a total of 50 player feedback data to their team coach, as well as one feedback from the team coach to the event organizer. Out of 10 teams, there are 10 team feedback data.

4. Conclusions

With the rapid development of modern information technology, an industry called "E-sports" has emerged. As an emerging and rapidly developing industry, while leveraging the advantages of the internet, it will also be negatively affected by the internet. A suitable management system is needed to manage event data and information. The system includes four major roles, each of which has functions that can meet the basic needs of esports data management. Firstly, the role of system administrator is one of its main responsibilities, which is the management of system roles. The player role has relatively limited permissions, mainly for querying relevant data and event arrangements. The coach's role requires attention and timely adjustment of the competitive status of the team's players. The main task of the event organizer in this system is to design competition details, arrange competition details, and publish them.

In terms of development technology, Spring Boot framework technology and Vue were adopted. JS technology is currently a popular technology with reliable stability and security. These technologies make development easier and enable developers to focus more on business logic. Including the MySQL database used, it also has the characteristics of small size and free use, making it more suitable for remote use.

The functions developed by this system temporarily meet the basic requirements of E-sports data management, and feedback and other functions have been added to data management. Subsequent related research can be conducted based on the implementation of system functions, by adding or improving functions to make the system more diverse, enhancing its usage range, optimizing user experience, such as enriching screen content, increasing injury ratio, attendance rate, and other data.

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